

共建开源生态

共建开源生态是我们的目标。为此我们努力把性价比和颜值做到极致。

同时全面开放SDK、原理图、PCB等所有软硬件资料。

提供交流平台，免费训练营、开发者扶持计划等，让您能够尽情发挥您的才华，

我们坚信自由和定制化是开源生态的核心，

泰山派开发板不仅仅是立创开发板的产品，它更是我们共同努力的成果。

我们只是抛砖引玉，期望能够激发出更多的创意和想法，

我们渴望与志同道合的伙伴们携手合作，共同探索无限可能。

让我们各展所长，共同建设一个强大的开源生态系统。

泰山派开发板等待着您的加入。

让我们一起推动技术的前进，一起创未来！



立创·泰山派介绍

泰山派是一款开源的卡片电脑，提供全面开放的软硬件资料，愿与志同道合的伙伴们共同推动技术的发展和创。

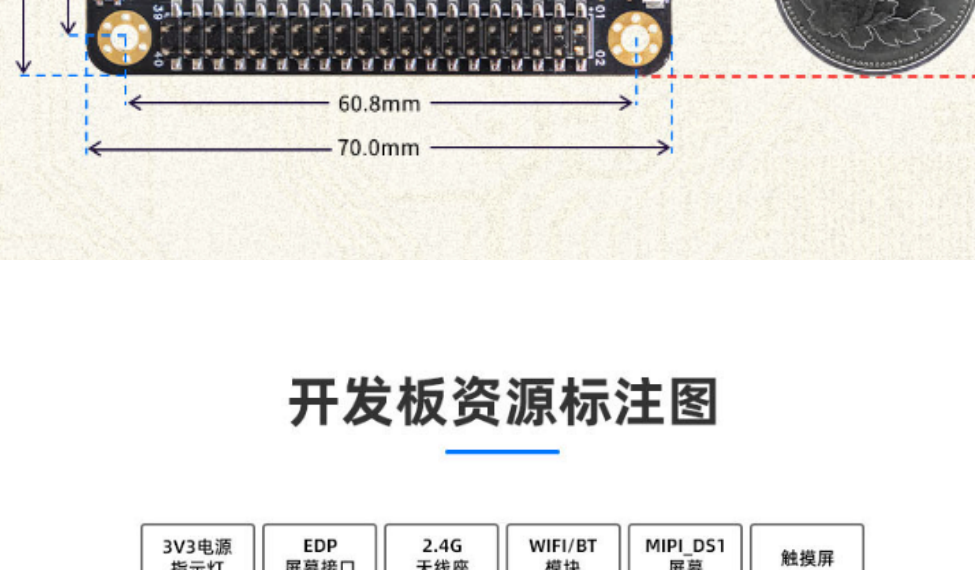
小巧的板子搭载了高配的处理器，引出丰富的外部资源、多样性SDK，赋予创意无限可能。

软硬件结合、项目式学习、解决项目落地难问题、让每一个想法变成现实。

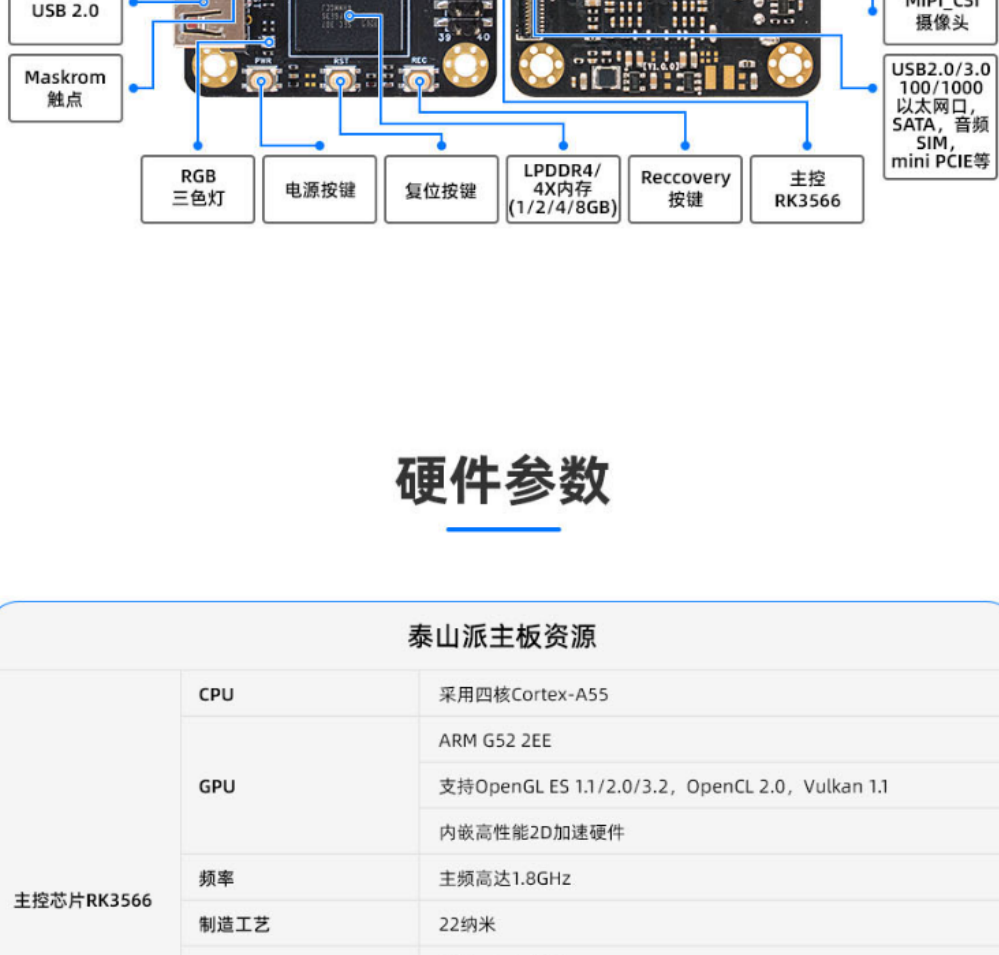


小到极致 强到别致

70mm X 45mm 比卡片更小的电脑
14个外设接口，500+个器件，79个拓展IO口



开发板资源标注图



硬件参数

泰山派主板资源		
主控芯片RK3566	CPU	采用自研Cortex-A55
	GPU	ARM G52 2EE
	频率	内置高性能2D加速硬件
	制造工艺	22nm
	NPU	高达1.0TOP算力
	视频解码	支持4K 60fps H.265/H.264/VP9
	视频编码	支持1080P 60fps H.265/H.264
	ISP	8M ISP HDR
	内存芯片	LPDDR4/4X
	存储芯片	EMMC
供电方式	USB_Type_C端子	5V3A输入
	GH1.25供电端子	5-18V，2A输入
	MICRO HDMI电源接口	1000m9120Hz，4K60Hz
	EDP屏幕接口	2560 X 1600@60Hz
	MIPI DSI屏幕接口	1080P@60Hz
	摄像头	MIPI CSI摄像头接口
	摄像头	30PIN 4Lane 摄像头
	通信网络接口	USB2.0端子
	WiFi/蓝牙模块	Type-A端子
	调试接口	频率:2.4GHz;发射功率:16dBm;接收灵敏度:-90dBm;
其他接口或模块	TF卡座	支持TF卡启动系统，最高支持512GB
	40Pin GPIO	兼容经典40Pin接口，支持PWM，I2C，SPI，UART，GPIO功能
	39Pin 高速信号扩展接口	USB2.0/3.0，100/1000以太网口，SATA，音频，SIM，mini PCIe等
	扬声器和麦克风接口	RK809-5内置1.3W 扬声器
	摄像头	RK809-5内置摄像头
	按键	RK809复位按键
	按键	系统复位按键
	按键	Recovery按键
	灯	RGB三色LED灯
	灯	RGB三色LED灯

主控、内存、存储

主控：Rockchip RK3566 是中高端AIoT芯片，采用2nm制程工艺，4核Cortex-A55的64位CPU，主频高达1.8GHz；集成ARM Mali-G52 GPU，支持4K 60fps解码，1080P 60fps编码，支持8M ISP和HDR；内置1TOPs算力的AI加速NPU；

内存：1/2/4/8GB可选；

存储：EMMC (B/16/32/64/128GB) & TF (最高支持512GB)



支持3屏显示+触摸



任意双屏同显



解锁AI潜力，开启智慧新篇章

高通的4Lane摄像头输入接口
RK3566内置的NPU 算力高达1TOPs,支持多种AI计算框架，包括TensorFlow Lite，Caffe，MXNet等。
快速处理人脸识别，动物识别，文字识别，垃圾识别等AI任务，让智能设备更加智能化。



板载WiFi/蓝牙

灵活搭配板载WiFi/BT模块
适配市面上各种WiFi/BT模块，确保兼容性，全方位满足各类应用需求。

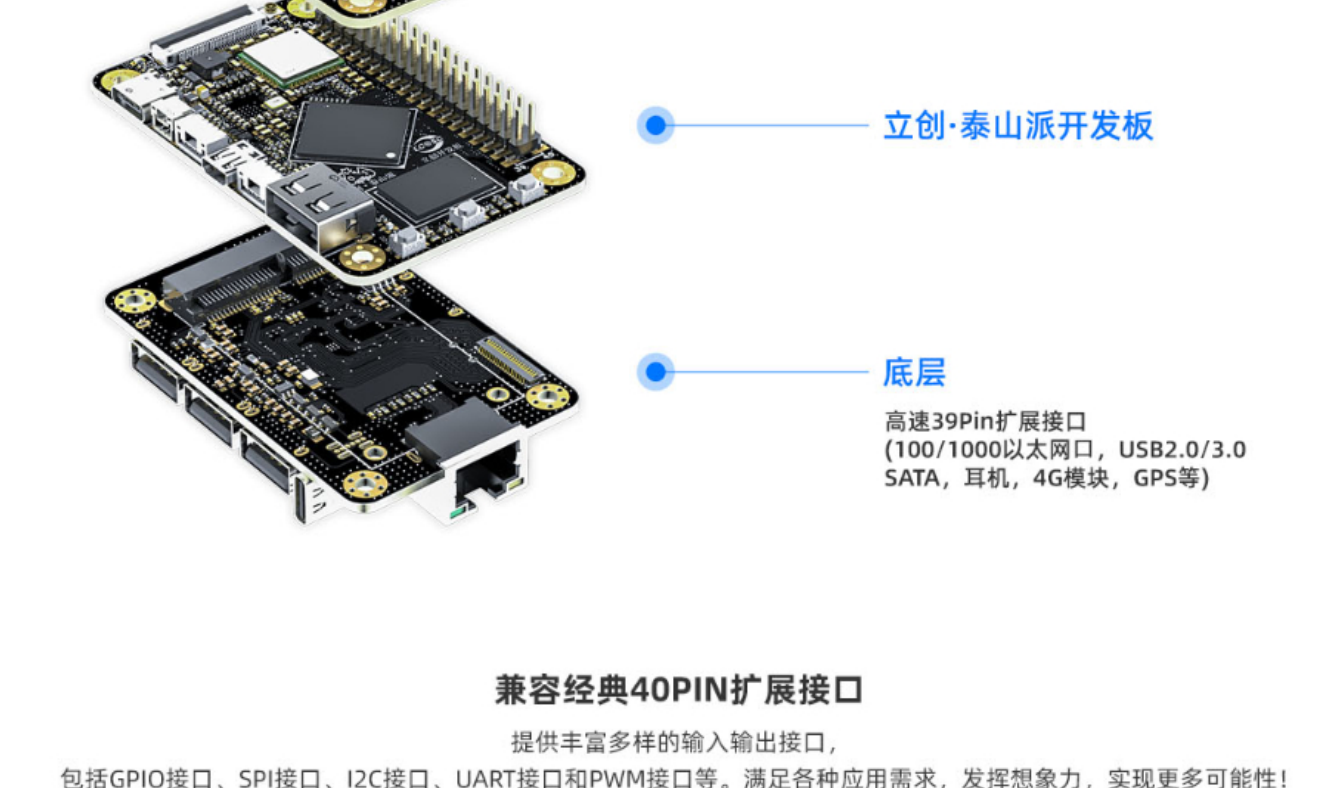


泰山派开发板为项目而生

兼容经典生态探索全新可能性



叠层设计，扩展无限可能



兼容经典40Pin扩展接口

提供丰富多样输入输出接口
包括GPIO接口，SPI接口，I2C接口，UART接口和PWM接口等，满足各种应用需求，发挥想象力，实现更多可能性！

引脚名称	GPIO	引脚名称	GPIO	引脚名称	GPIO
GPIO_0	GPIO_0	GPIO_1	GPIO_1	GPIO_2	GPIO_2
GPIO_3	GPIO_3	GPIO_4	GPIO_4	GPIO_5	GPIO_5
GPIO_6	GPIO_6	GPIO_7	GPIO_7	GPIO_8	GPIO_8
GPIO_9	GPIO_9	GPIO_10	GPIO_10	GPIO_11	GPIO_11
GPIO_12	GPIO_12	GPIO_13	GPIO_13	GPIO_14	GPIO_14
GPIO_15	GPIO_15	GPIO_16	GPIO_16	GPIO_17	GPIO_17
GPIO_18	GPIO_18	GPIO_19	GPIO_19	GPIO_20	GPIO_20
GPIO_21	GPIO_21	GPIO_22	GPIO_22	GPIO_23	GPIO_23
GPIO_24	GPIO_24	GPIO_25	GPIO_25	GPIO_26	GPIO_26
GPIO_27	GPIO_27	GPIO_28	GPIO_28	GPIO_29	GPIO_29
GPIO_30	GPIO_30	GPIO_31	GPIO_31	GPIO_32	GPIO_32
GPIO_33	GPIO_33	GPIO_34	GPIO_34	GPIO_35	GPIO_35
GPIO_36	GPIO_36	GPIO_37	GPIO_37	GPIO_38	GPIO_38
GPIO_39	GPIO_39	GPIO_40	GPIO_40	GPIO_41	GPIO_41

丰富高速接口，助您大胆创新

众多扩展接口支持定义，100/1000以太网，USB2.0/3.0，MINI PCIe，SATA，GPS，耳机，GPIO接口等等

引脚名称	GPIO	引脚名称	GPIO	引脚名称	GPIO
GPIO_0	GPIO_0	GPIO_1	GPIO_1	GPIO_2	GPIO_2
GPIO_3	GPIO_3	GPIO_4	GPIO_4	GPIO_5	GPIO_5
GPIO_6	GPIO_6	GPIO_7	GPIO_7	GPIO_8	GPIO_8
GPIO_9	GPIO_9	GPIO_10	GPIO_10	GPIO_11	GPIO_11
GPIO_12	GPIO_12	GPIO_13	GPIO_13	GPIO_14	GPIO_14
GPIO_15	GPIO_15	GPIO_16	GPIO_16	GPIO_17	GPIO_17
GPIO_18	GPIO_18	GPIO_19	GPIO_19	GPIO_20	GPIO_20
GPIO_21	GPIO_21	GPIO_22	GPIO_22	GPIO_23	GPIO_23
GPIO_24	GPIO_24	GPIO_25	GPIO_25	GPIO_26	GPIO_26
GPIO_27	GPIO_27	GPIO_28	GPIO_28	GPIO_29	GPIO_29
GPIO_30	GPIO_30	GPIO_31	GPIO_31	GPIO_32	GPIO_32
GPIO_33	GPIO_33	GPIO_34	GPIO_34	GPIO_35	GPIO_35
GPIO_36	GPIO_36	GPIO_37	GPIO_37	GPIO_38	GPIO_38
GPIO_39	GPIO_39	GPIO_40	GPIO_40	GPIO_41	GPIO_41

项目式学习

软硬件结合，通过亲身参与需求分析、硬件设计、程序编写，到3D外壳设计，全方位体验项目诞生的全过程，掌握项目式开发。解决项目落地难问题，让每一个想法变成现实



支持多种操作系统



广泛的应用场景



产品展示



售后服务

售后服务承诺：

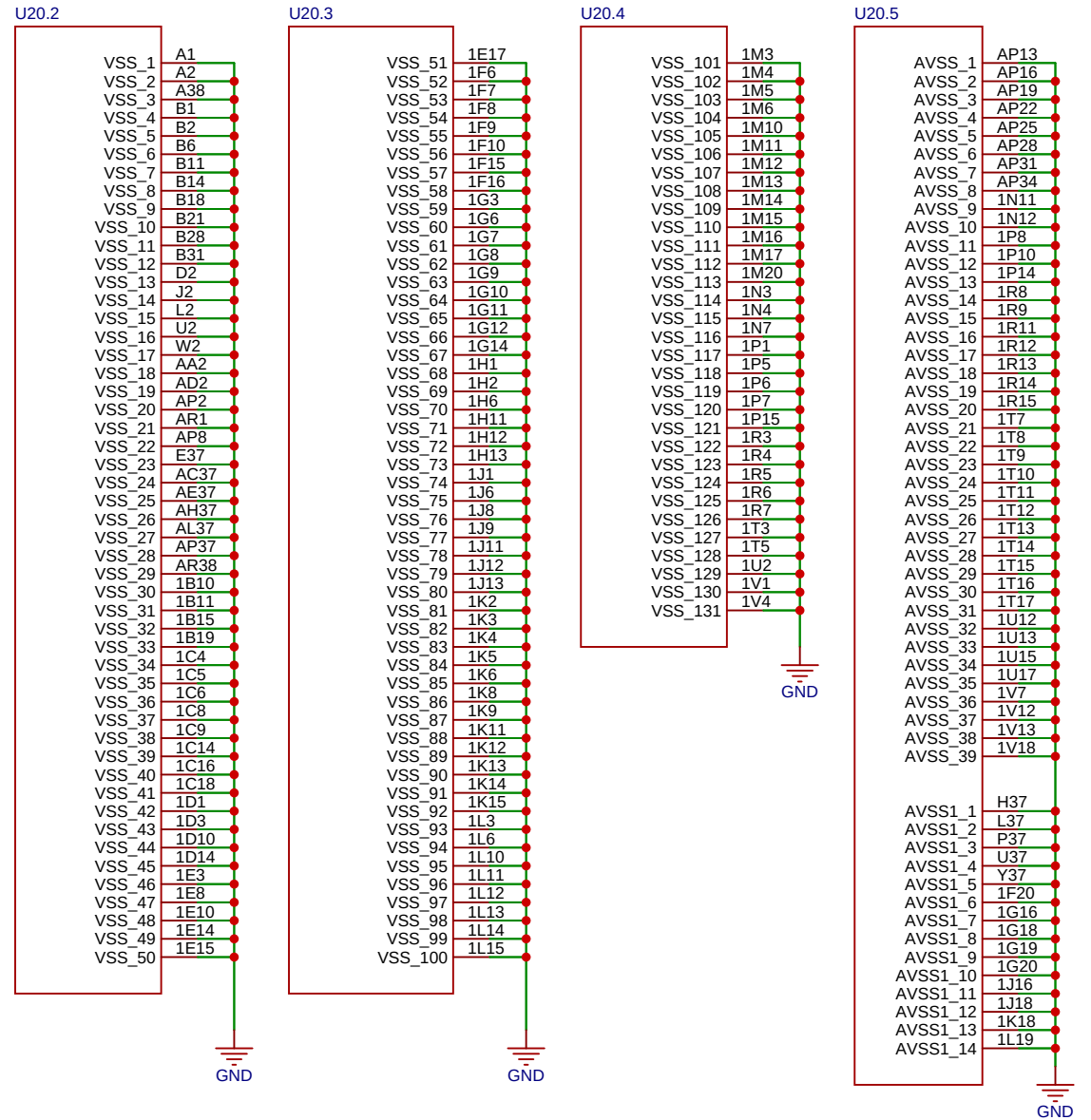
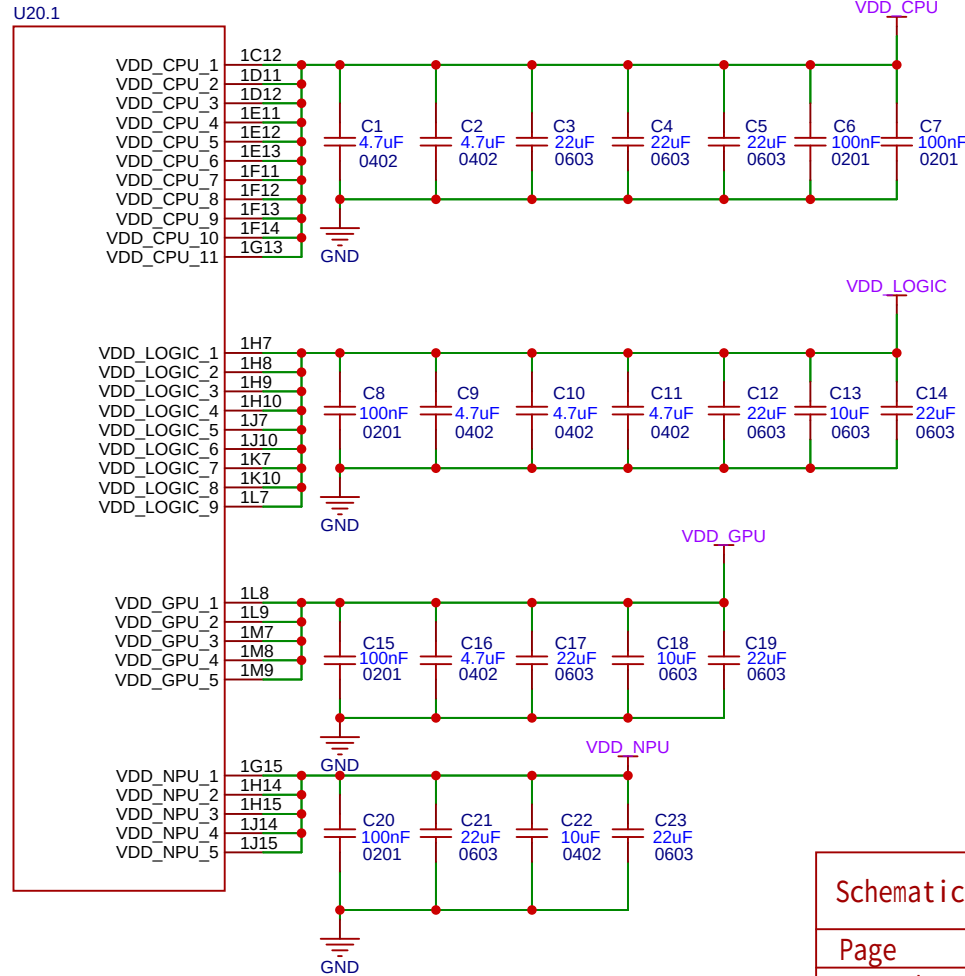
- 不支持退货
- 1个月内：没有维修费，产品问题我们负责来回运费换新或维修。
- 人为因素损坏，买家需要支付物料成本使用以及来回运费，不收取维修费。
- 1个月以后：产品问题或人为损坏，买家需承担来回运费和物料成本使用。
- 不收取维修手续费。使用过程中如果损坏严重，例如插接导致焊盘脱落，则维修不了。

联系方式



RK3566_12345

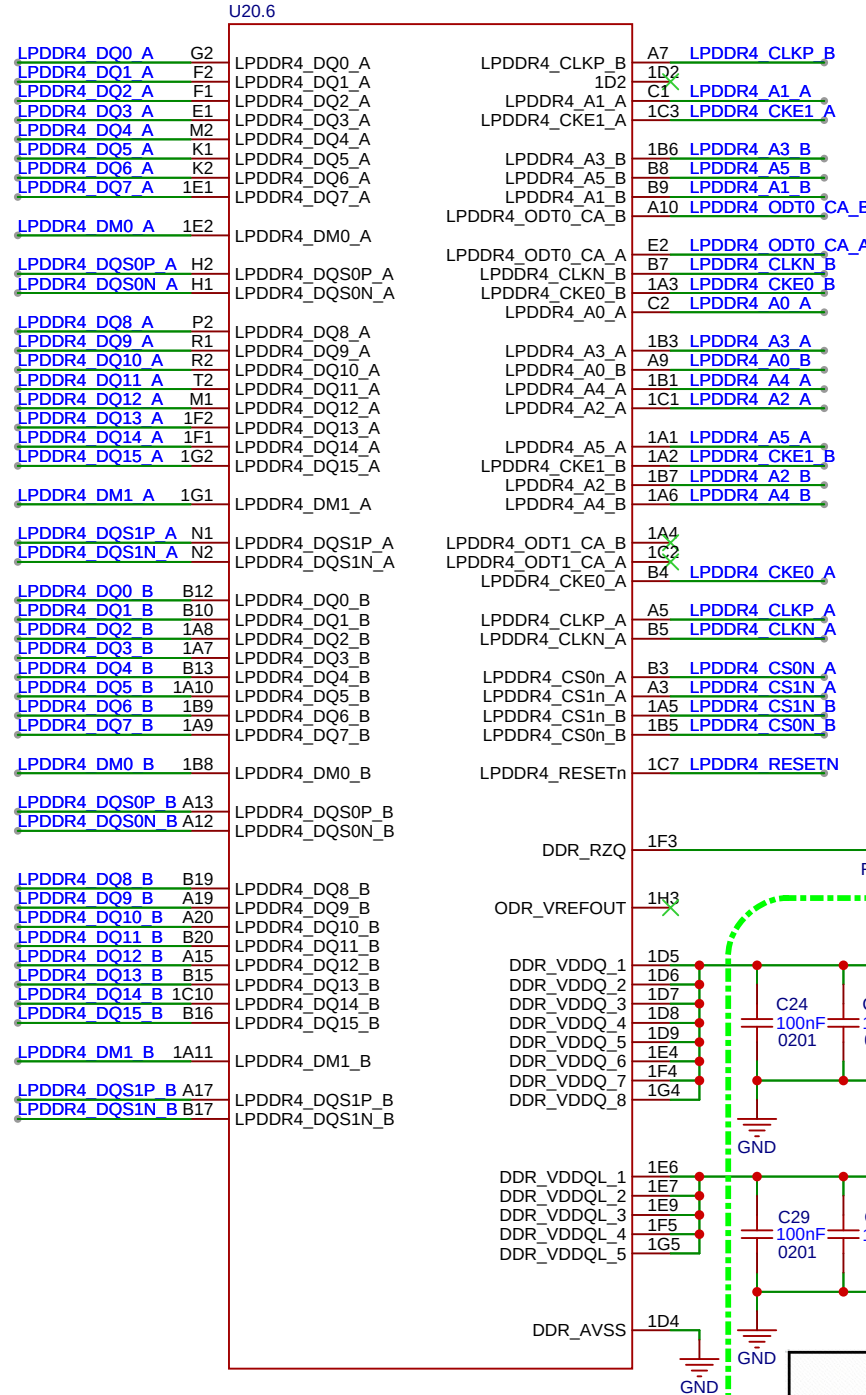
(POWER)



Schematic	Schematic	Update Date	
		Create Date	
Page	RK3566_POWER&GND		Part Number
Drawn	立创开发板	【内部】立创·泰山派开发板	
Reviewed			
		VER	SIZE
		V0.0.3	A4
		PAGE	3 OF 29
		立创EDA	

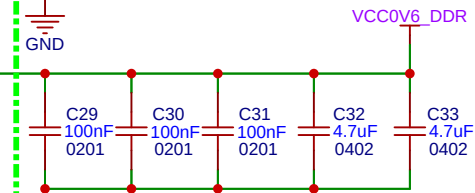
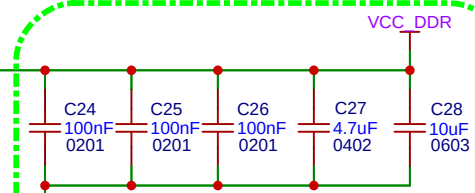
RK3566_6

(DDR_PHY)



VCC_DDR

R1 R0402 120Ω ±1% The resistor parameter is 120R ±1%



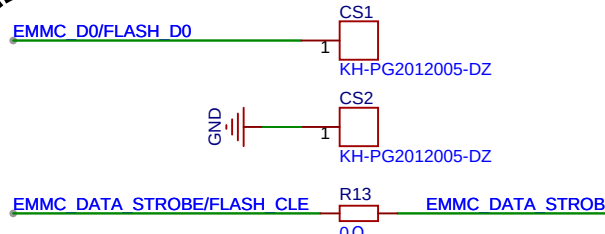
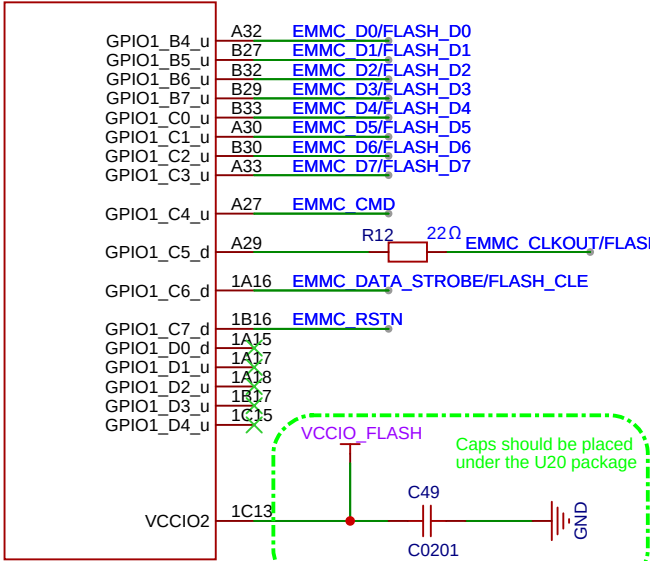
Caps should be placed under the U20 package

	VDDQ	VDDQL
LPDDR4	1.10V	1.10V
LPDDR4X	1.10V	0.60V

(OSC/PLL/PMUIO1/2)



RK3566_9
(VCCIO2 Domain)

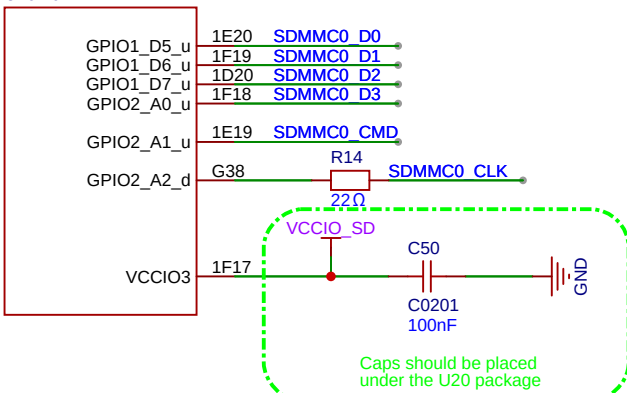


eMMC Short 'eMMC_D0' and GND to enter Maskrom Mode.

Be sure to reserve the testpoints for
firmware burning or update.

软件dts的电压配置严格与硬件设计保持一致
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

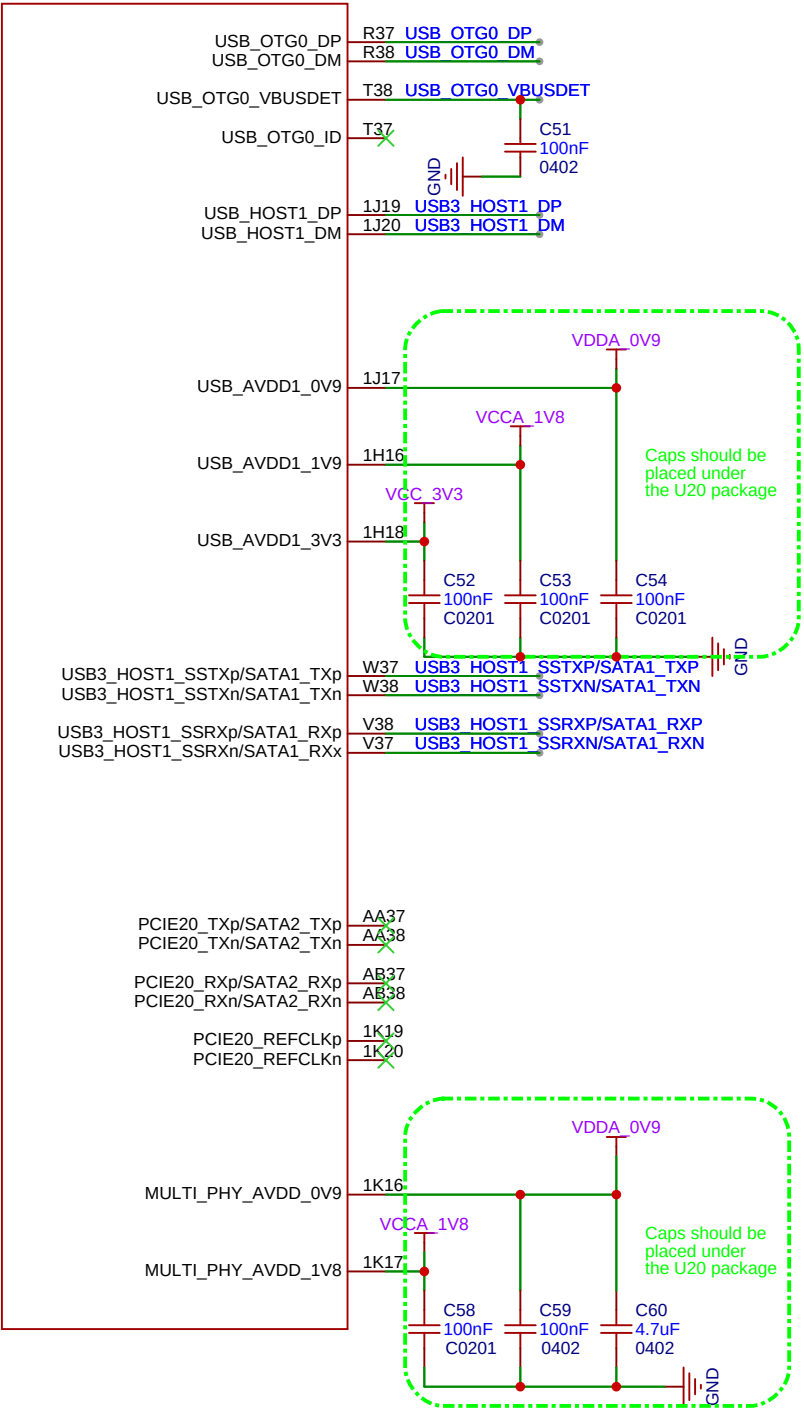
RK3566_10
(VCCIO3 Domain)



软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

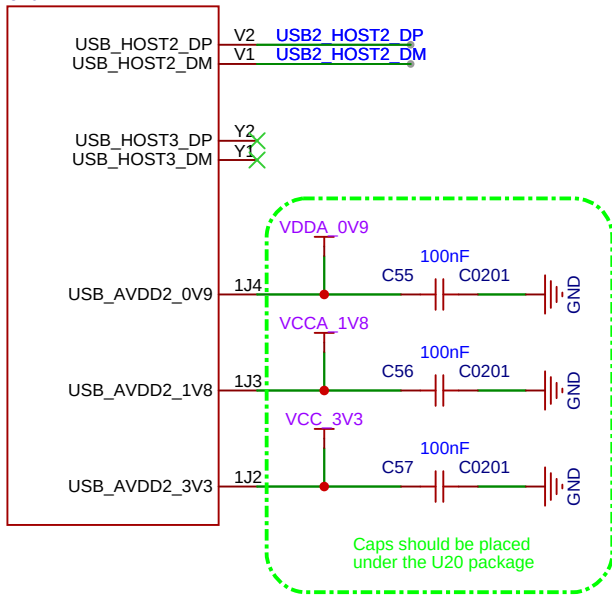
RK3566_21(USB3.0/PCIe2.0x1/SATA)

U20.21



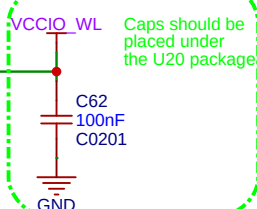
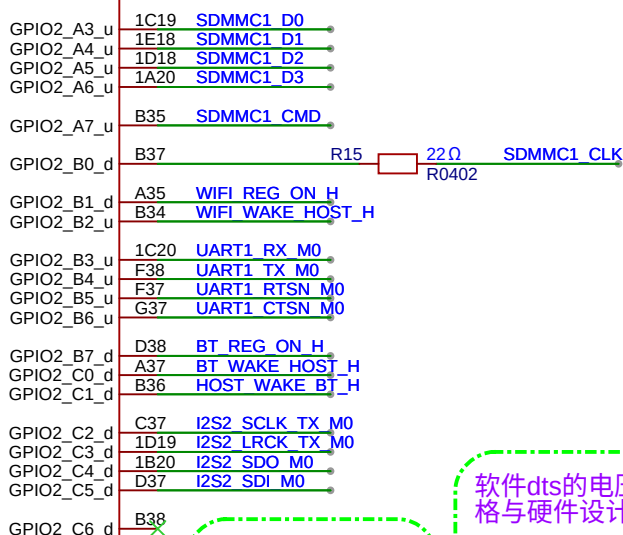
RK3566_22
(USB2.0 HOST)

U20.22



RK3566_11 (VCCIO4 Domain)

U20.11

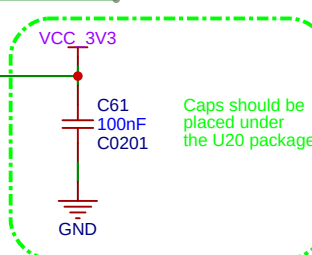
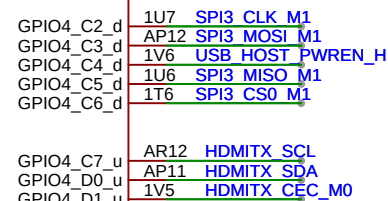


软件dts的电压配置严格与硬件设计保持一致！

Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.

RK3566_14 (VCCIO7 Domain)

U20.14

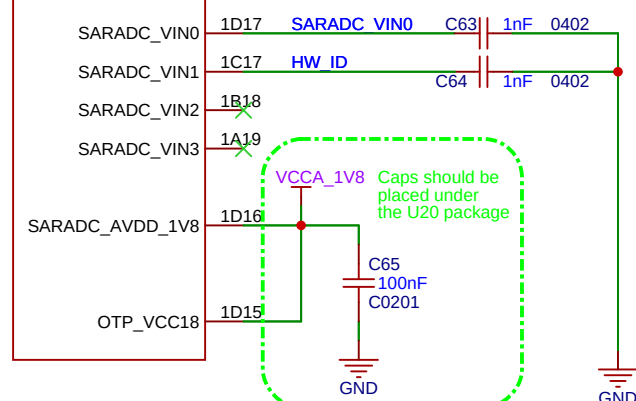


软件dts的电压配置严格与硬件设计保持一致！

Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.

RK3566_15 (SARADC/OTP)

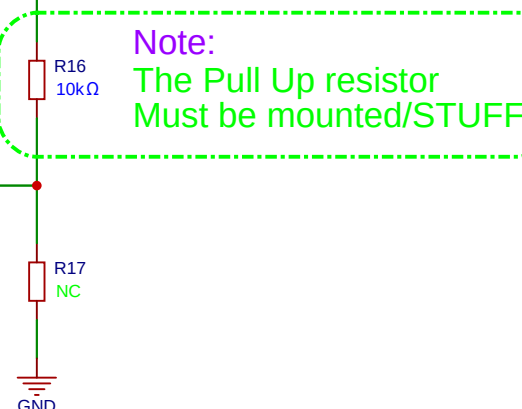
U20.15



Hardware board ID

HW ID

VCCA_1V8



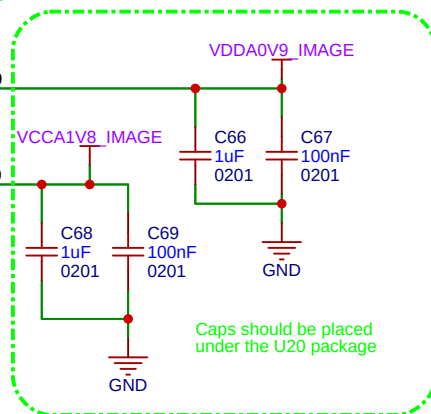
RK3566_16 (MIPI_CSI_RX)

U20.16

MIPI_CSI_RX_D0p AP18 MIPI CSI DP0
 MIPI_CSI_RX_D0n AR18 MIPI CSI DN0
 MIPI_CSI_RX_D1p AR17 MIPI CSI DP1
 MIPI_CSI_RX_D1n AP17 MIPI CSI DN1
 MIPI_CSI_RX_D2p AP15 MIPI CSI DP2
 MIPI_CSI_RX_D2n AR15 MIPI CSI DN2
 MIPI_CSI_RX_D3p AR14 MIPI CSI DP3
 MIPI_CSI_RX_D3n AP14 MIPI CSI DN3
 MIPI_CSI_RX_CLK0p 1V9 MIPI CSI MCP
 MIPI_CSI_RX_CLK0n 1U9 MIPI CSI MCN
 MIPI_CSI_RX_CLK1p 1V8
 MIPI_CSI_RX_CLK1n 1U8

MIPI_CSI_RX_AVDD_0V9

MIPI_CSI_RX_AVDD_1V8



RK3566_13 (VCCIO6 Domain)

U20.13

GPIO3_C6_d 1T2
 GPIO3_C7_d 1U1
 GPIO3_D0_d AP1
 GPIO3_D1_d AN2
 GPIO3_D2_d AM2
 GPIO3_D3_d AM1
 GPIO3_D4_d 1R2
 GPIO3_D5_d 1T1
 GPIO3_D6_d AL2 PHY TXD2
 GPIO3_D7_d AL1 PHY TXD3
 GPIO4_A0_d AK2 PHY TXCLK
 GPIO4_A1_d 1P2 PHY RXD2
 GPIO4_A2_d 1R1 PHY RXD3
 GPIO4_A3_d AJ2 PHY RXCLK
 GPIO4_A4_d 1N2 PHY TXD0
 GPIO4_A5_d AJ1 PHY TXD1
 GPIO4_A6_d AH2 PHY TXEN
 GPIO4_A7_d 1N1 PHY RXD0
 GPIO4_B0_d 1L2 PHY RXD1
 GPIO4_B1_d AG2 PHY RXDV
 GPIO4_B2_d AF2 I2C4 M0 SDA_CSI
 GPIO4_B3_d AE2 I2C4 M0 SCL_CSI
 GPIO4_B4_d AF1 PWDN_CSI
 GPIO4_B5_d AD1 RESET_CSI
 GPIO4_B6_d 1L1 PHY MDC
 GPIO4_B7_d AC2 PHY MDIO
 GPIO4_C0_d AB2 MCLK_CSI
 GPIO4_C1_d AB1 PHY MCLK

VCC_1V8

R18

I2C4 M0 SDA_CSI

R0402
 2.2kΩ ±1% 62.5mW

R19

I2C4 M0 SCL_CSI

R0402
 2.2kΩ ±1% 62.5mW

VCC_1V8

VCCIO6

VCCIO6_1

VCCIO6_2

1L4

1L5

R20

0Ω

C70

100nF

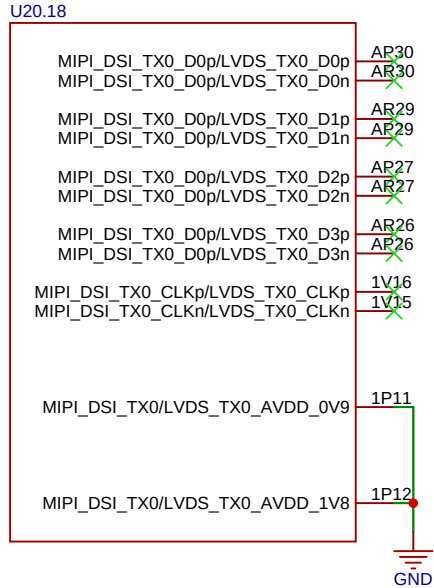
0201

GND

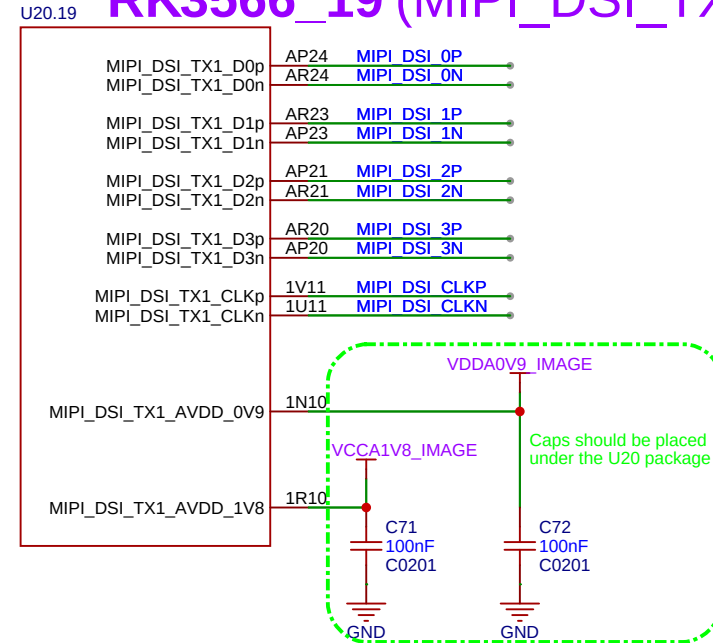
Caps should be placed under the U20 package

软件dts的电压配置严格与硬件设计保持一致！
 Check the software configuration(dts) of voltage level, which must be keep the same as hardware design.

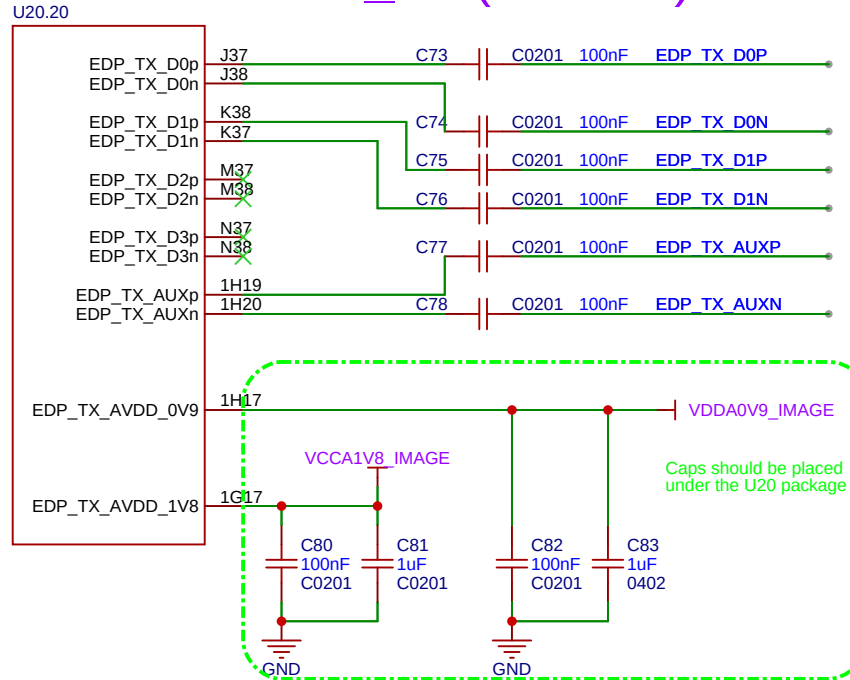
RK3566_18 (MIPI_DSI_TX0/LVDS_TX0)



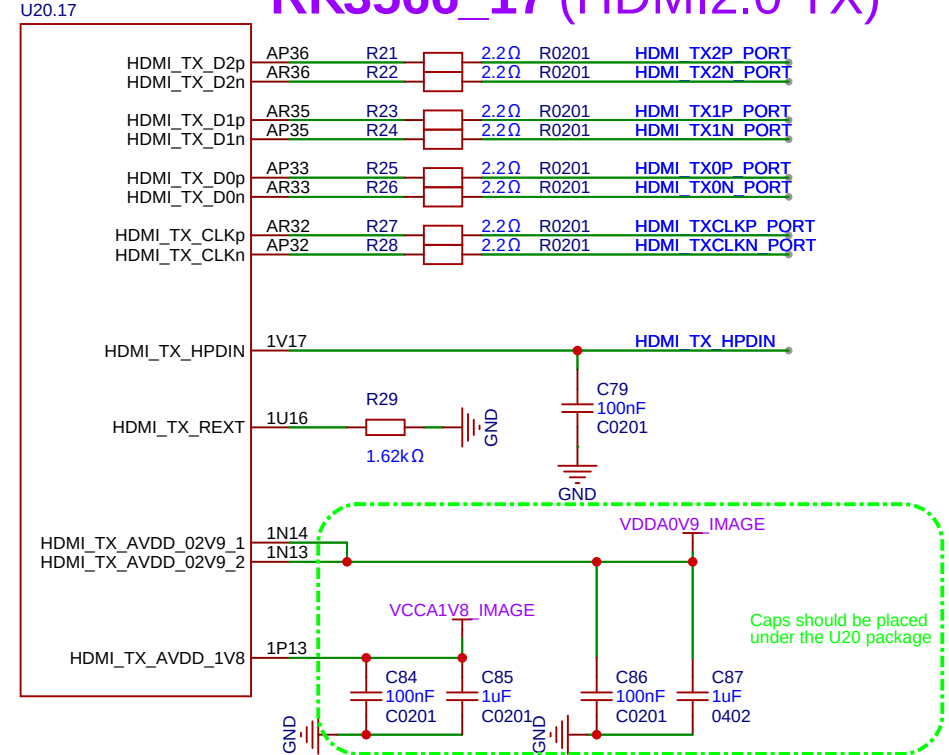
RK3566_19 (MIPI_DSI_TX1)



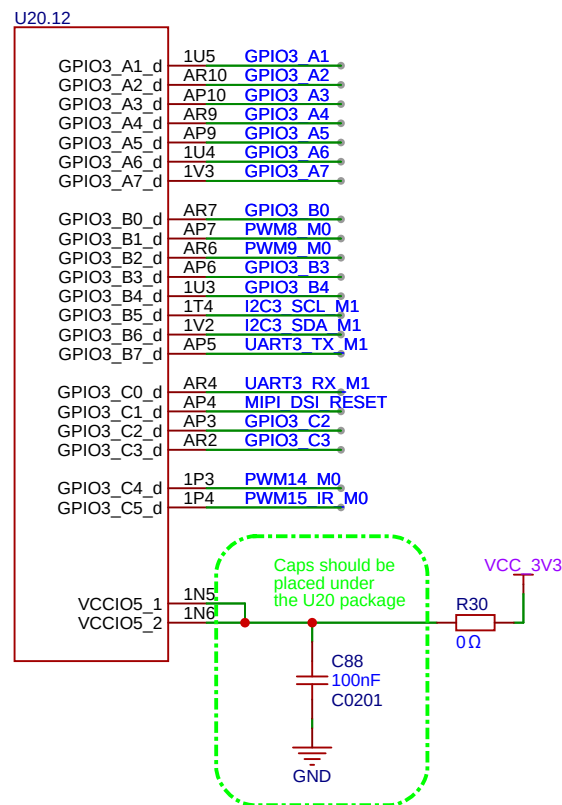
RK3566_20 (eDP TX)



RK3566_17 (HDMI2.0 TX)

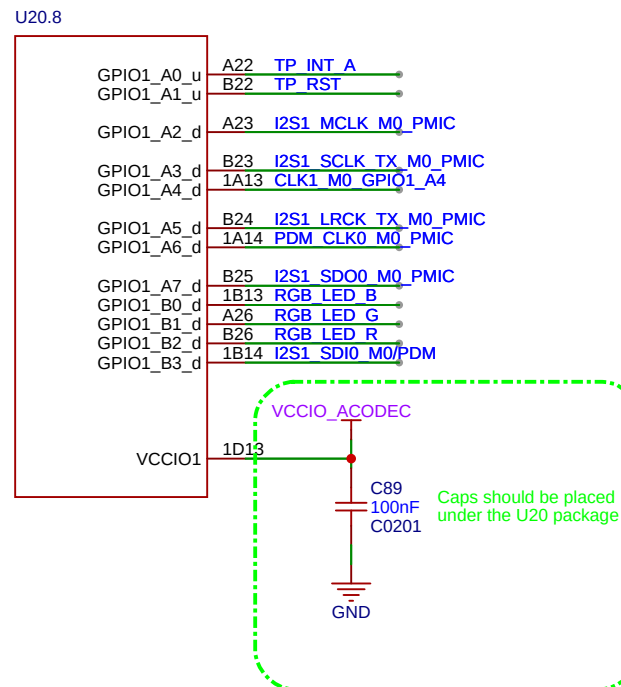


RK3566_12 (VCCIO5 Domain)



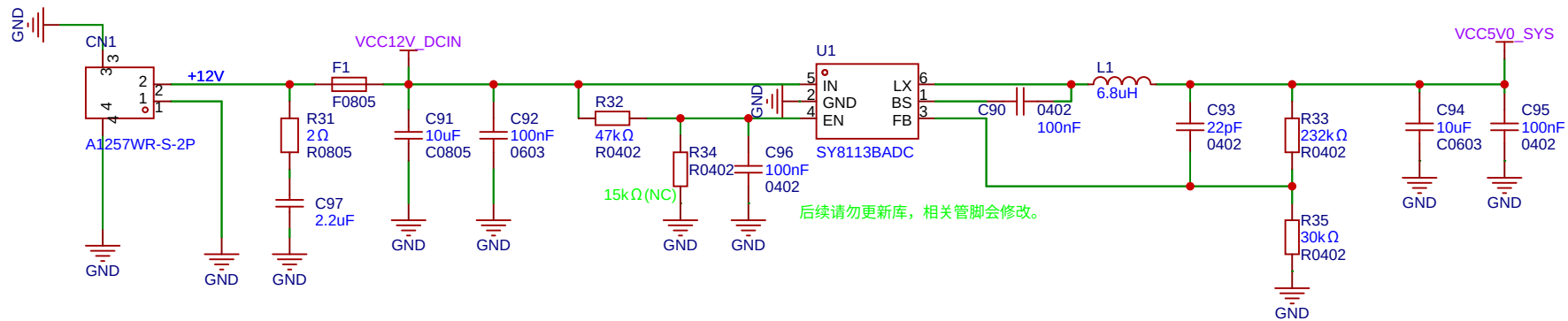
软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

RK3566_8 (VCCIO1 Domain)

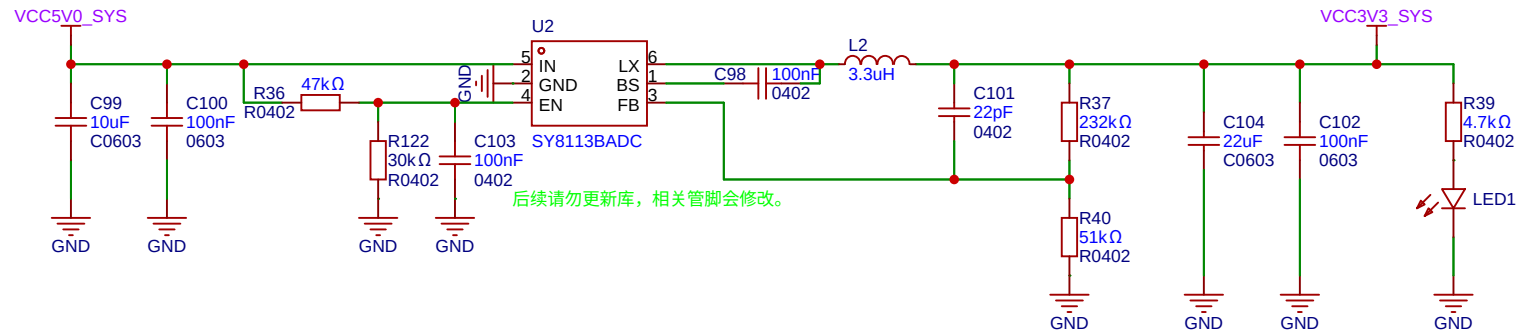


软件dts的电压配置严格与硬件设计保持一致！
Check the software configuration(dts)
of voltage level, which must be
keep the same as hardware design.

VCC 5-18V宽电压_DCIN 转 VCC_5V



VCC5V_SYS 转 VCC3V3_SYS



Schematic	Schematic		Update Date	
			Create Date	
Page	POWER		Part Number	
Drawn		【内部】立创·泰山派开发板		
Reviewed				
		VER	SIZE	PAGE 13 OF 29
嘉立创EDA		V1.0	A4	嘉立创EDA

[illegible]

VCCIO_WL

VCC_1V8

VCC_3V3

R55

R56

0Ω

R0402

NC

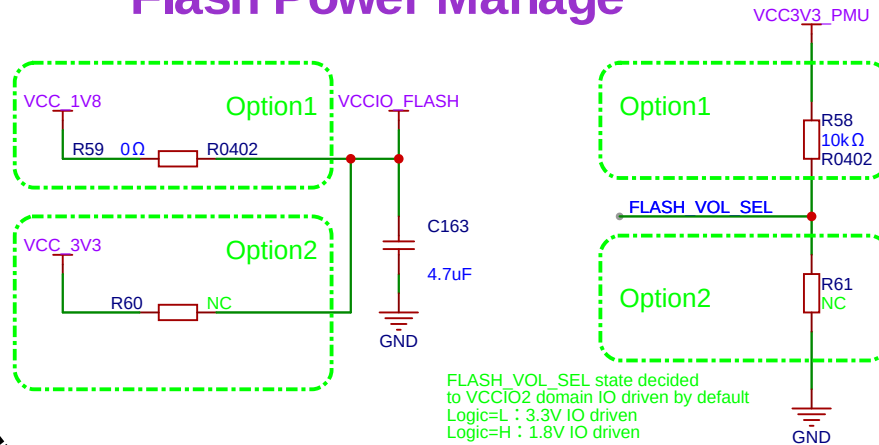
VCCIO_WL

C162

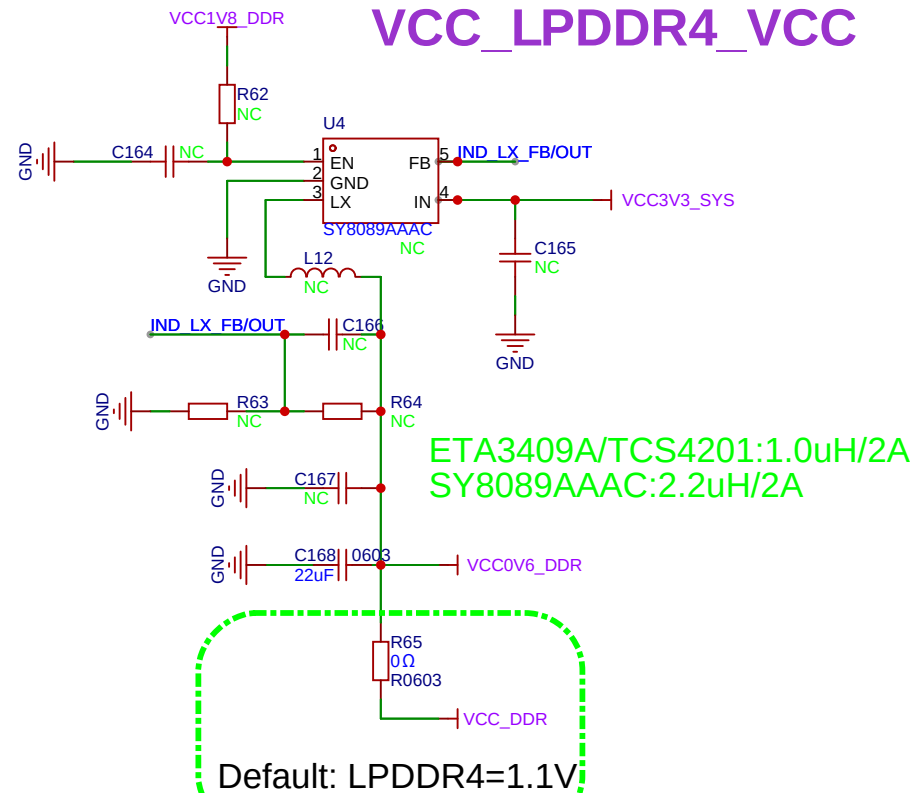
1μF

0402

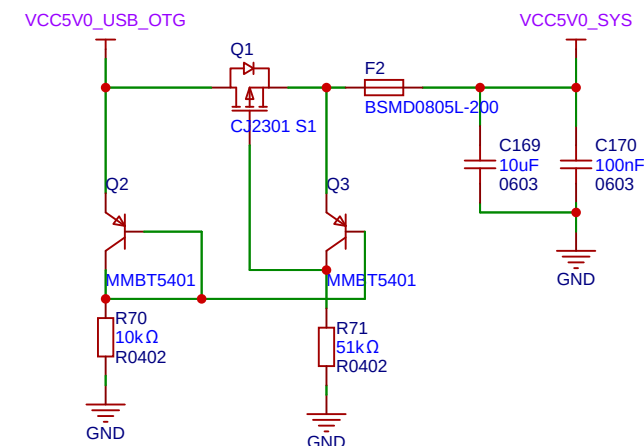
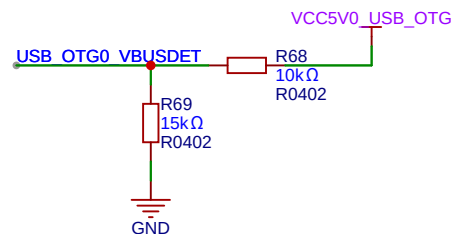
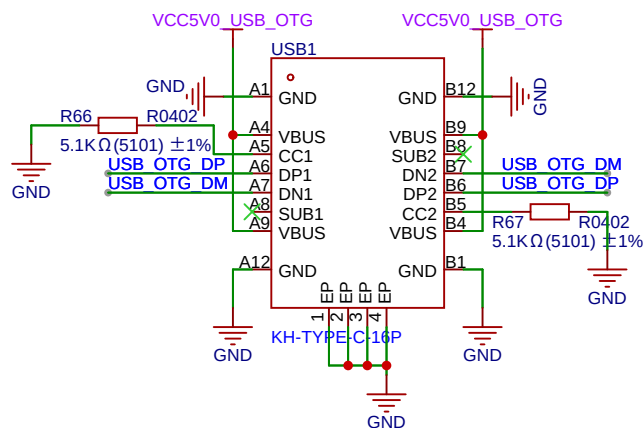
Flash Power Manage



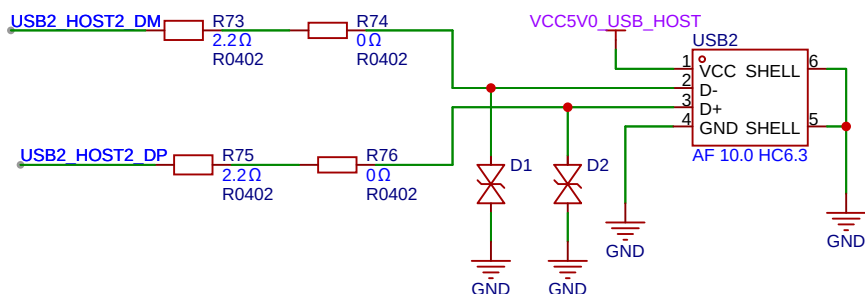
VCC_LPDDR4_VCC



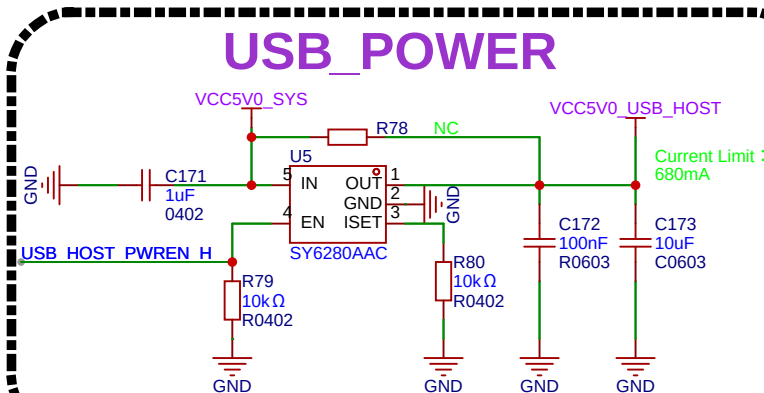
Power selection



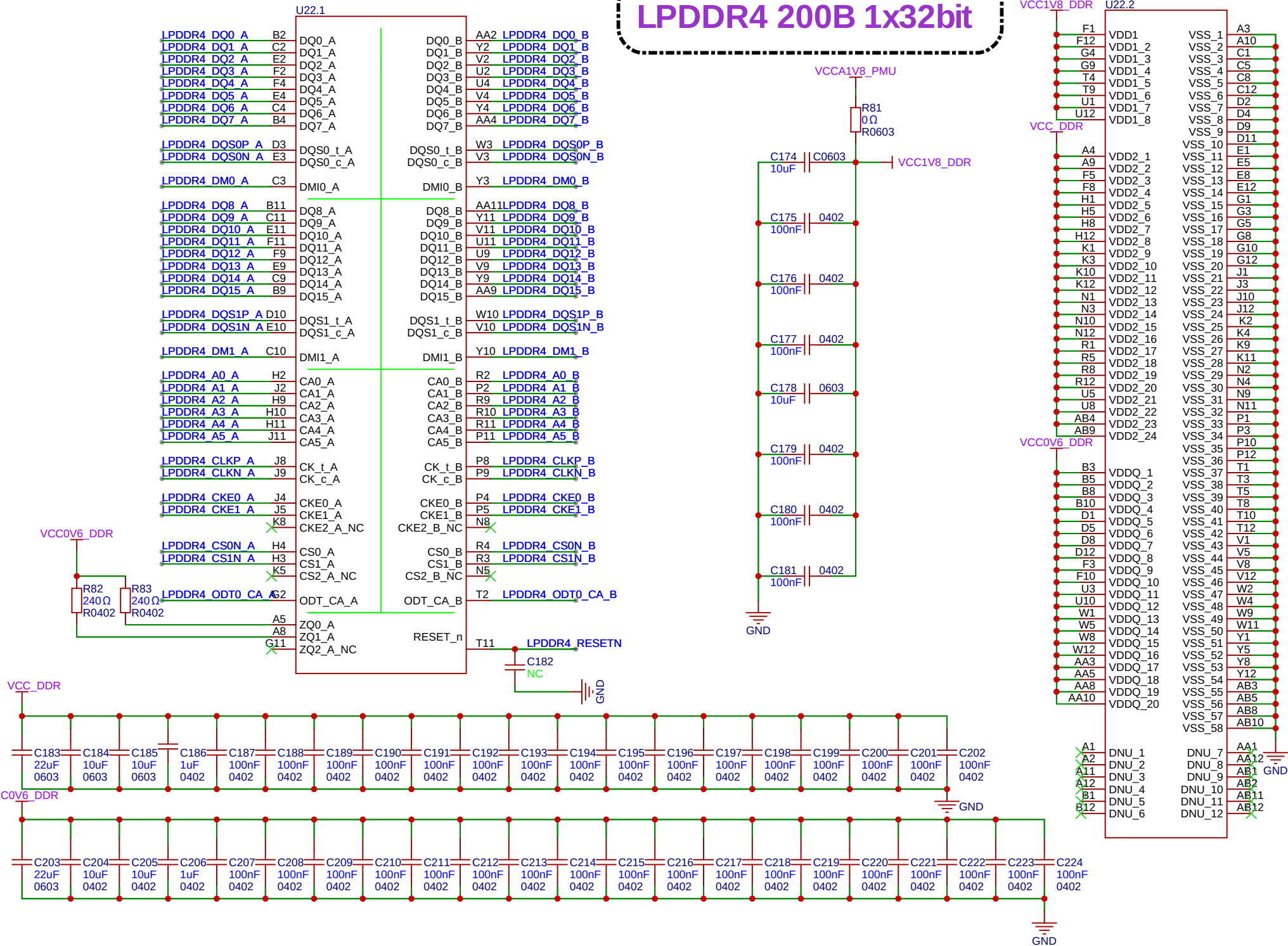
The diagram shows two signal lines: USB OTG DP and USB OTG DM. The DP line is connected to USB OTG0 DP through a 2.2Ω resistor (R72) in a 0402 package. The DM line is connected to USB OTG0 DM through a 2.2Ω resistor (R77) in a 0402 package. Both lines have diodes D3 and D4 connected to ground.



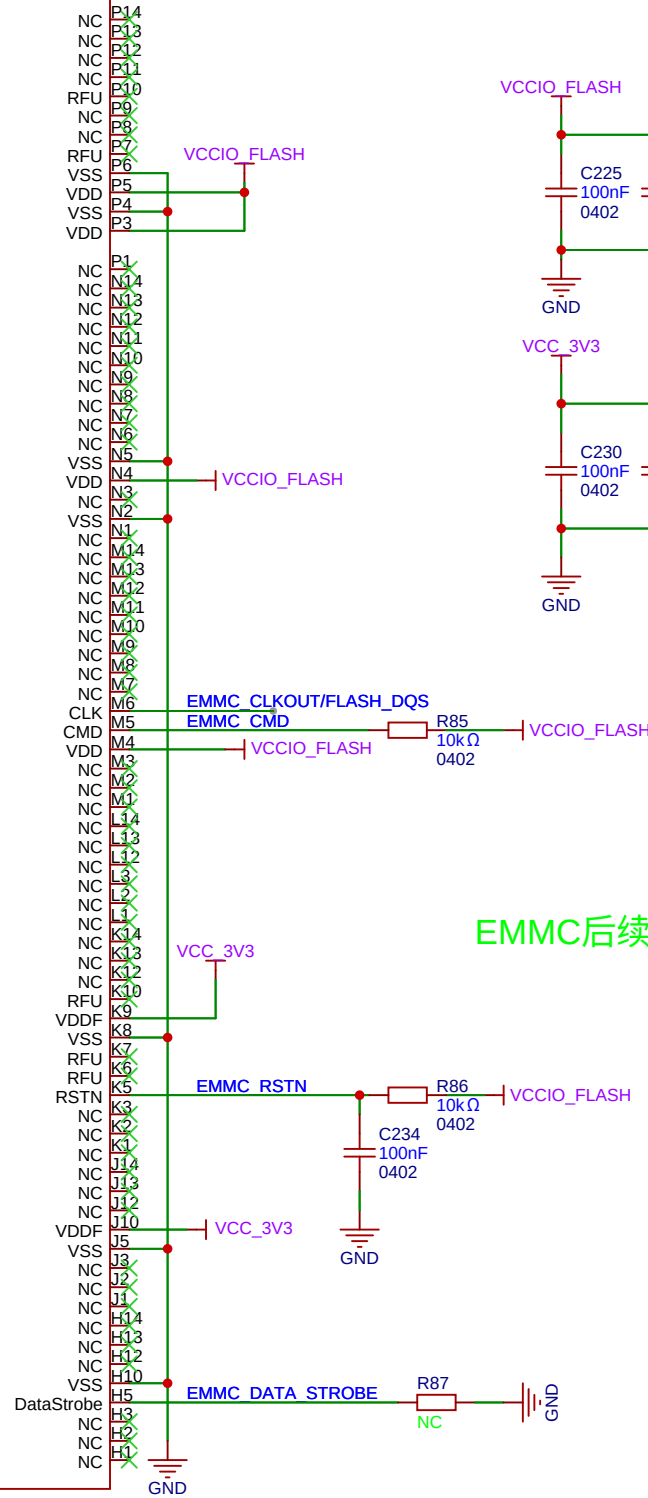
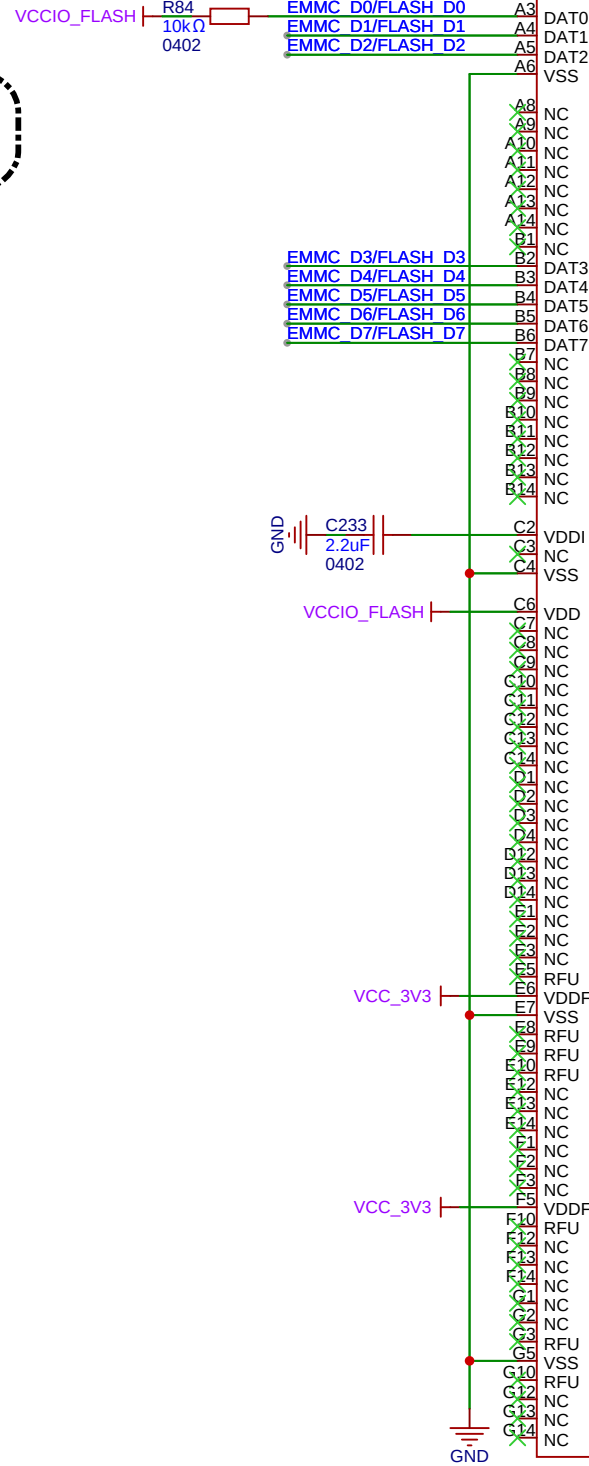
嘉立创EDA



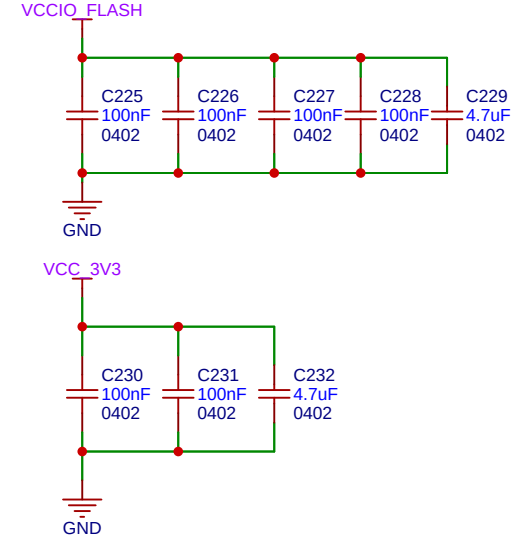
LPDDR4 200B 1x32bit



EMMC

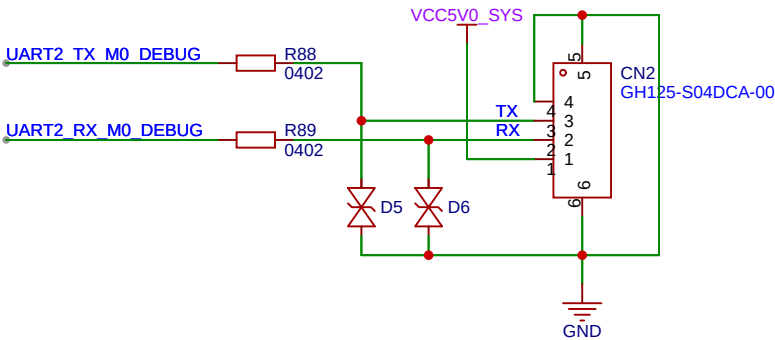


EMMC后续请勿更新库，相关管脚会修改。



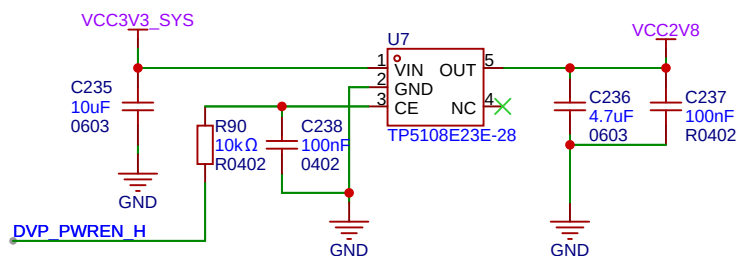
Debug

default 3.3V

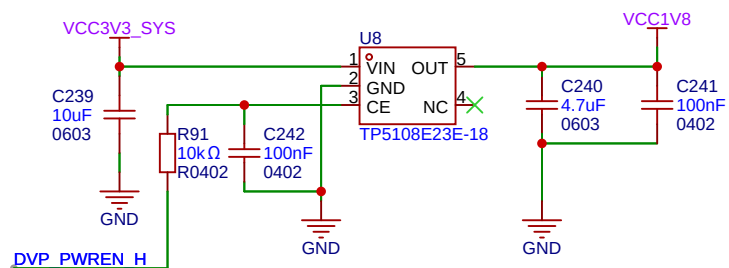


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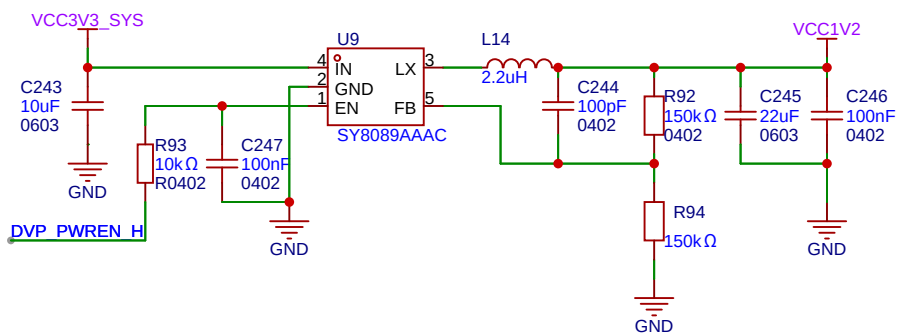
3V3转2V8



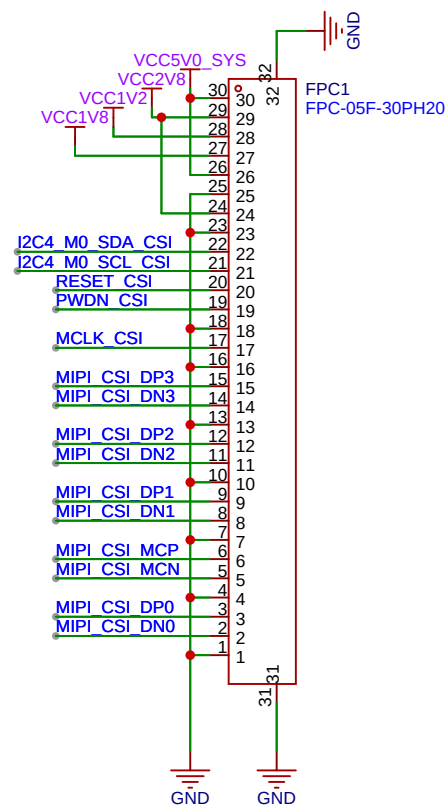
3V3转1V8

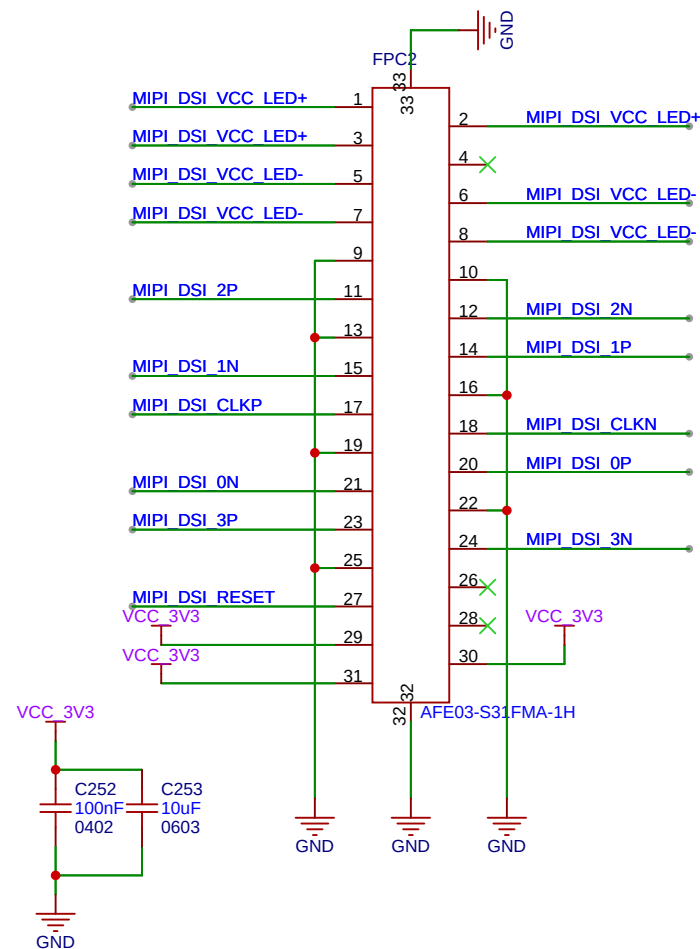


3V3转1V2

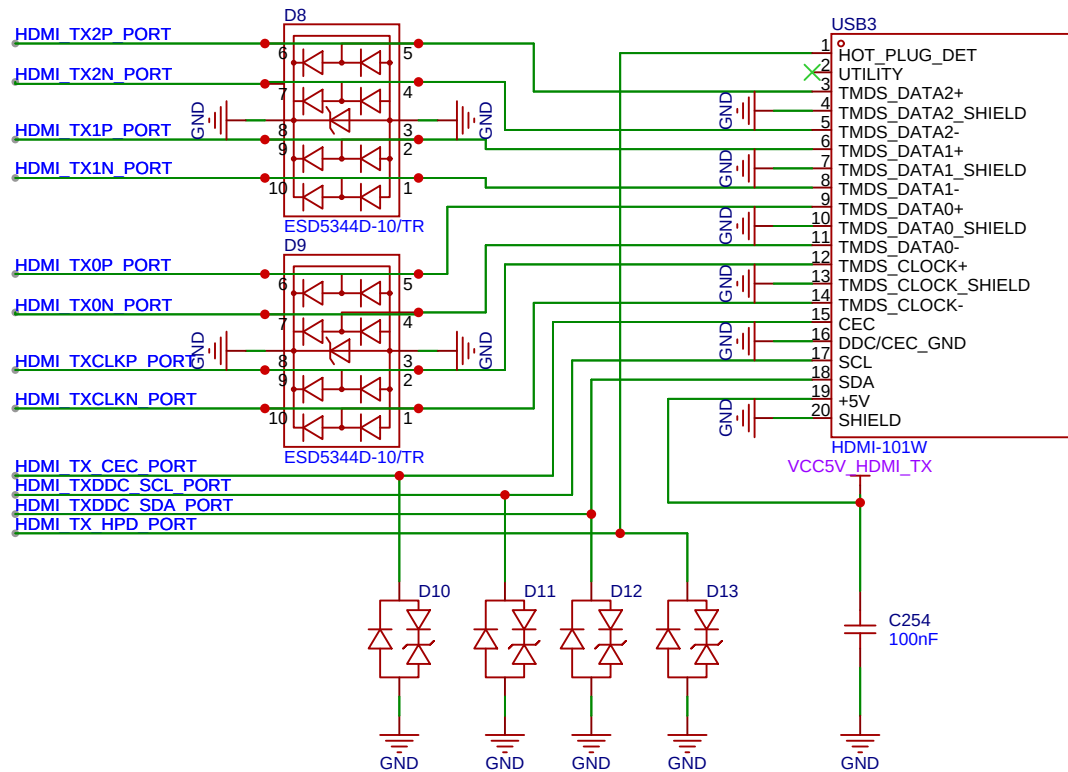


MIPI Camera (MIPI_CSI_RX 4lanes)

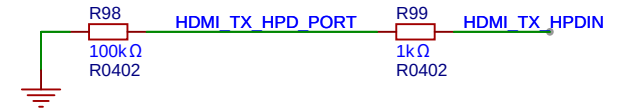




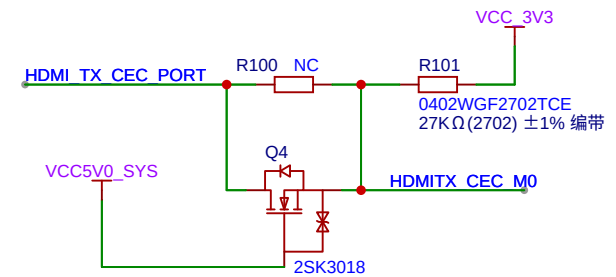
micro HDMI



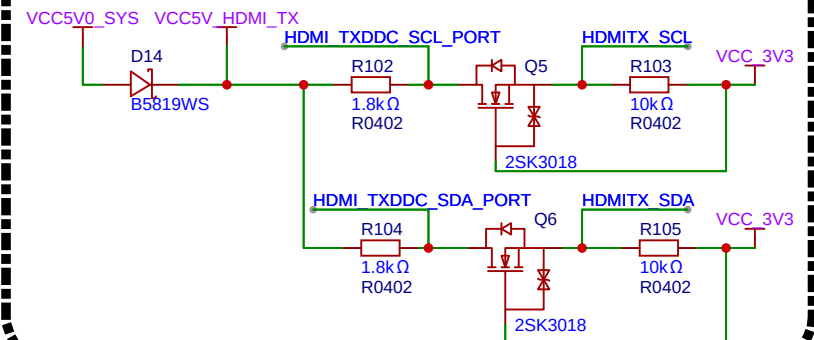
HDMI_TX_HPDP



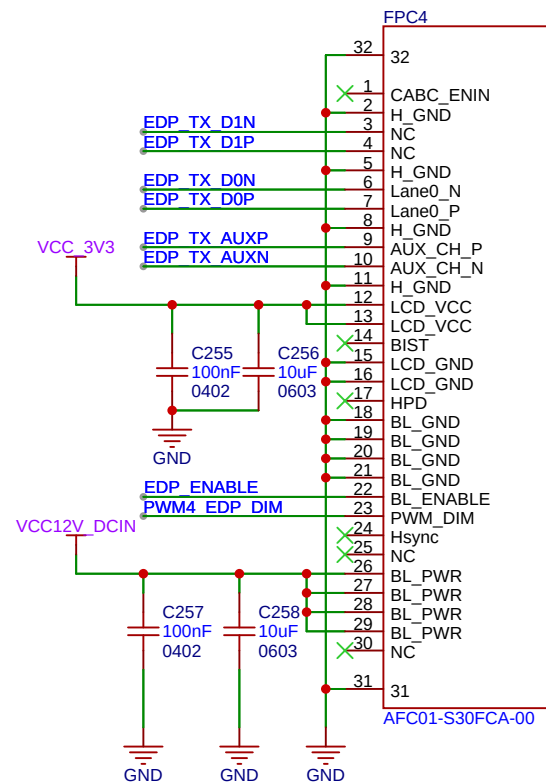
HDMI_TX_CEC



HDMI_TX_CEC

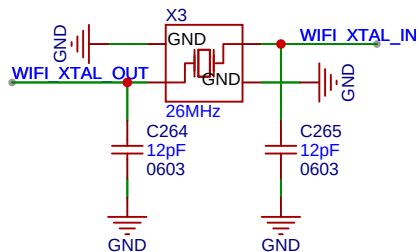


EDP_Panel

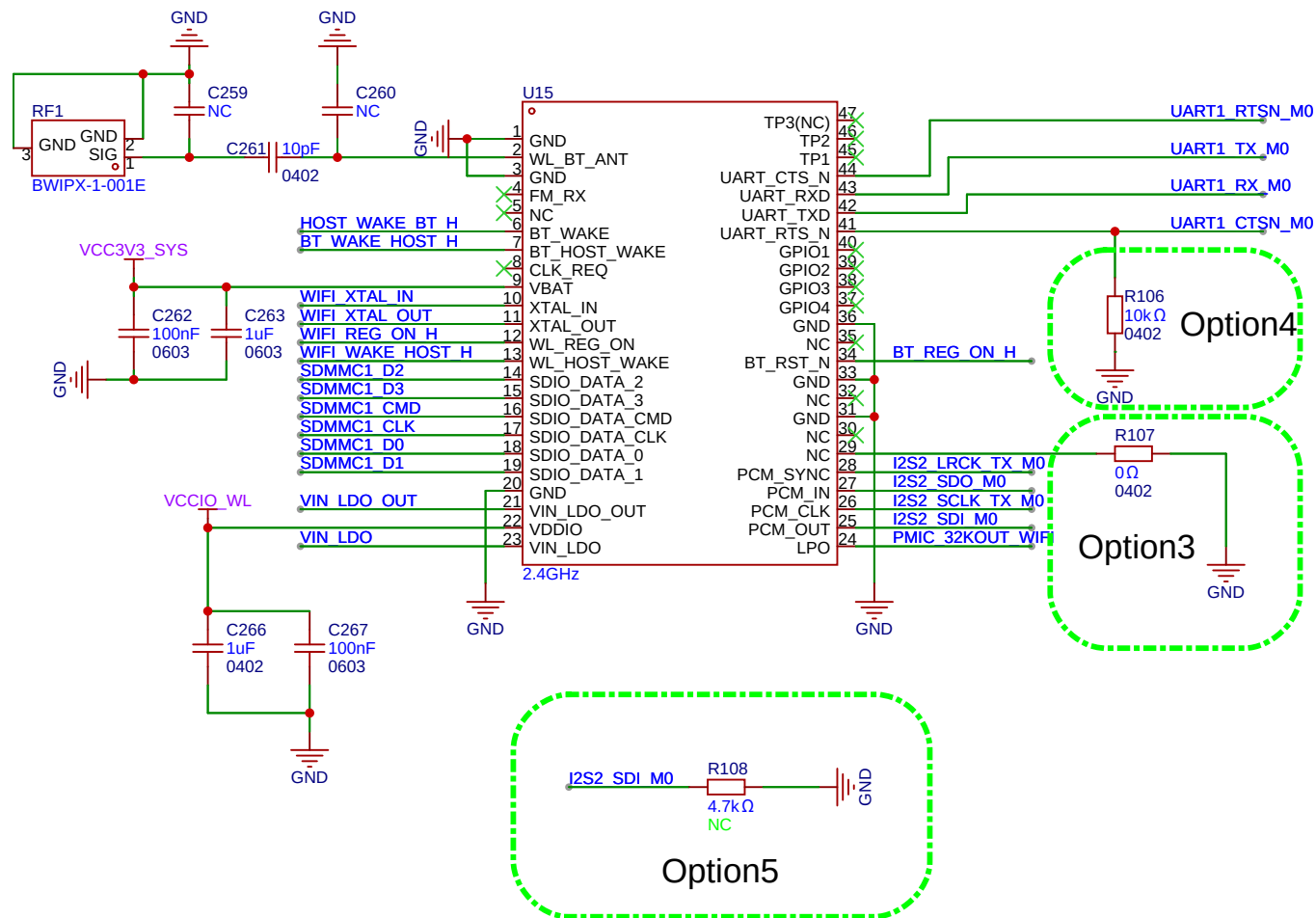
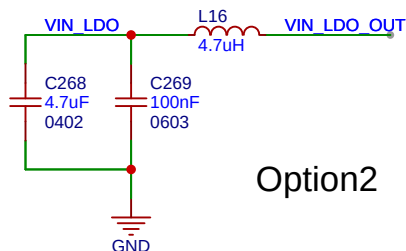


SDIO WIFI/BT

Option1



Option2



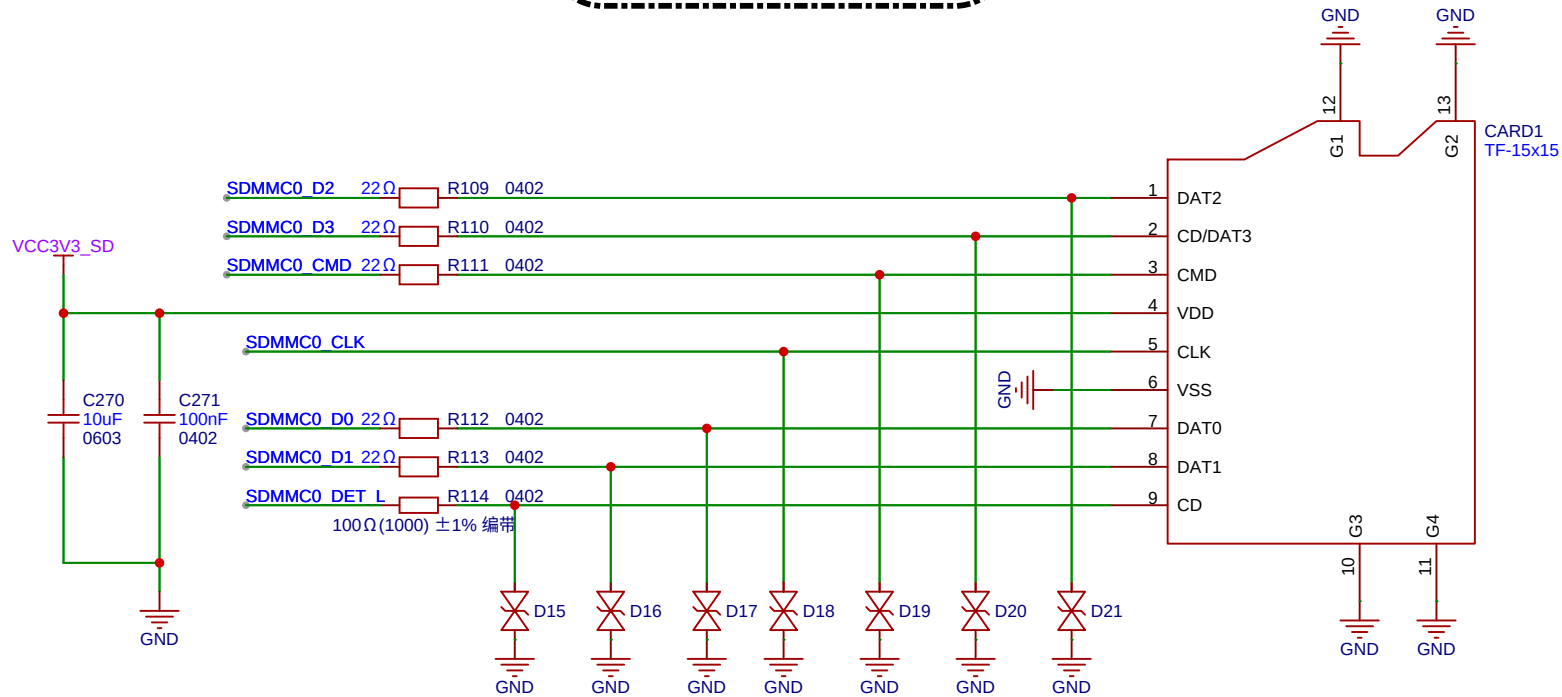
Option4

Option3

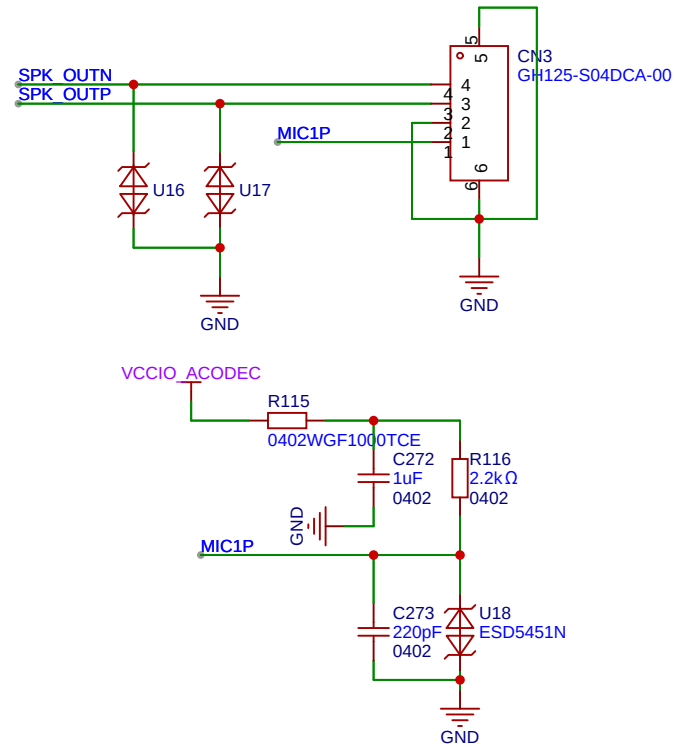
Option5

OPTION	WIFI				BT	Crystals	VCCIO_SDIO	OPTION1	OPTION2	OPTION3	OPTION4	OPTION5
	a	b/g/n	ac	5GHz								
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	No	No
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No	No
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	Yes	Yes
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No	No

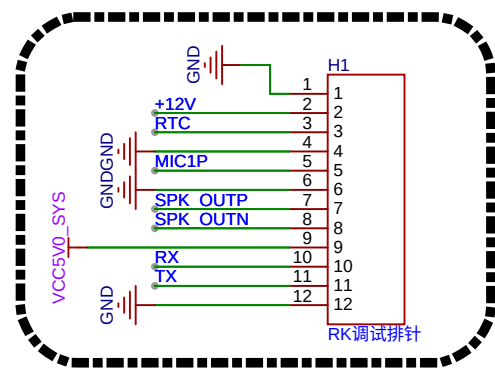
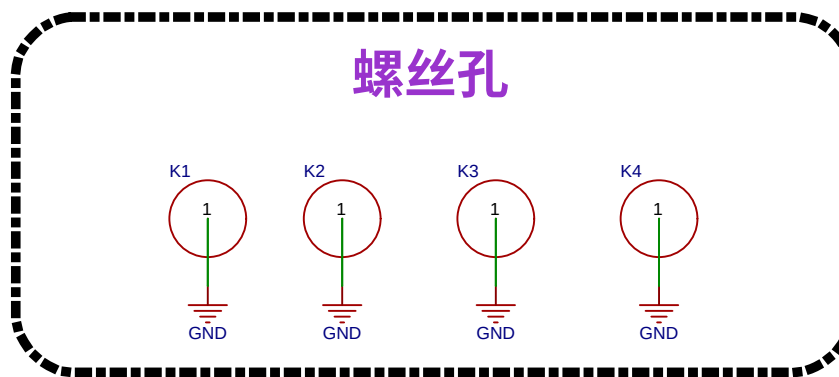
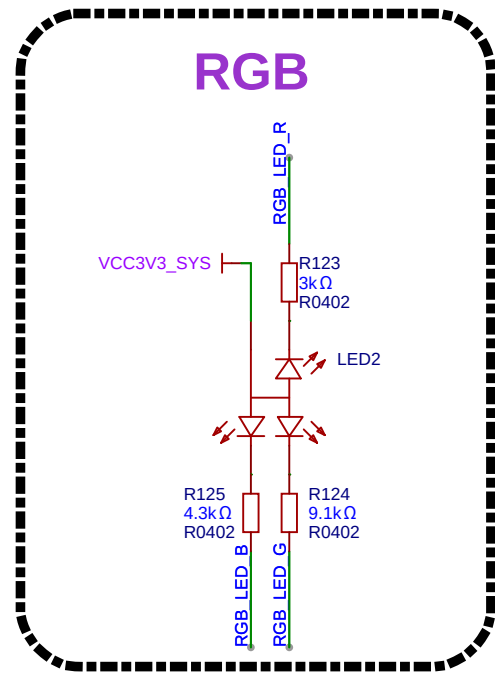
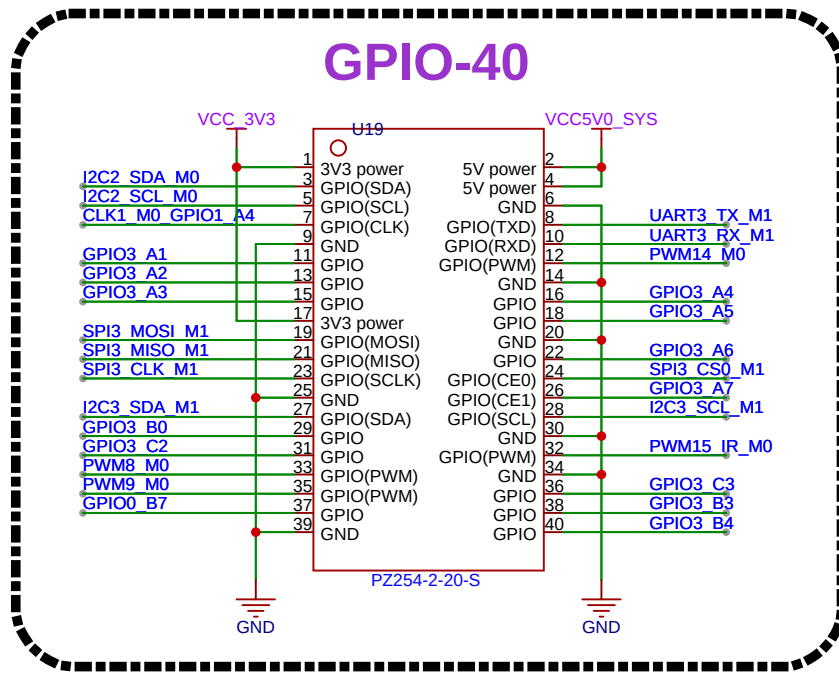
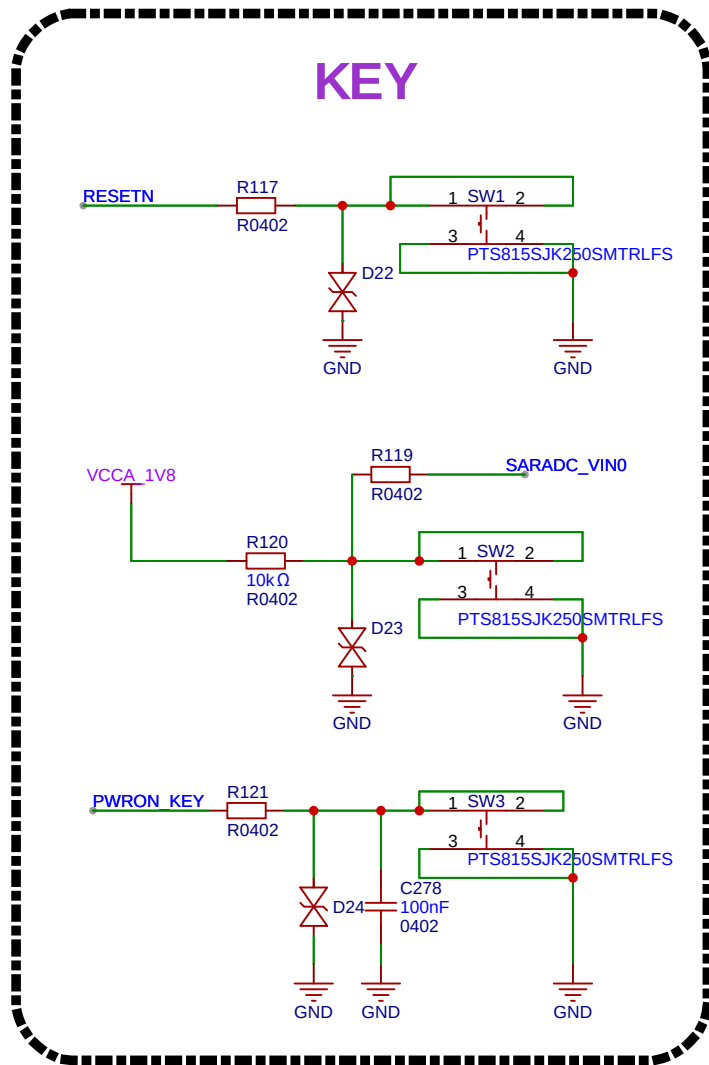
MicroSD Card



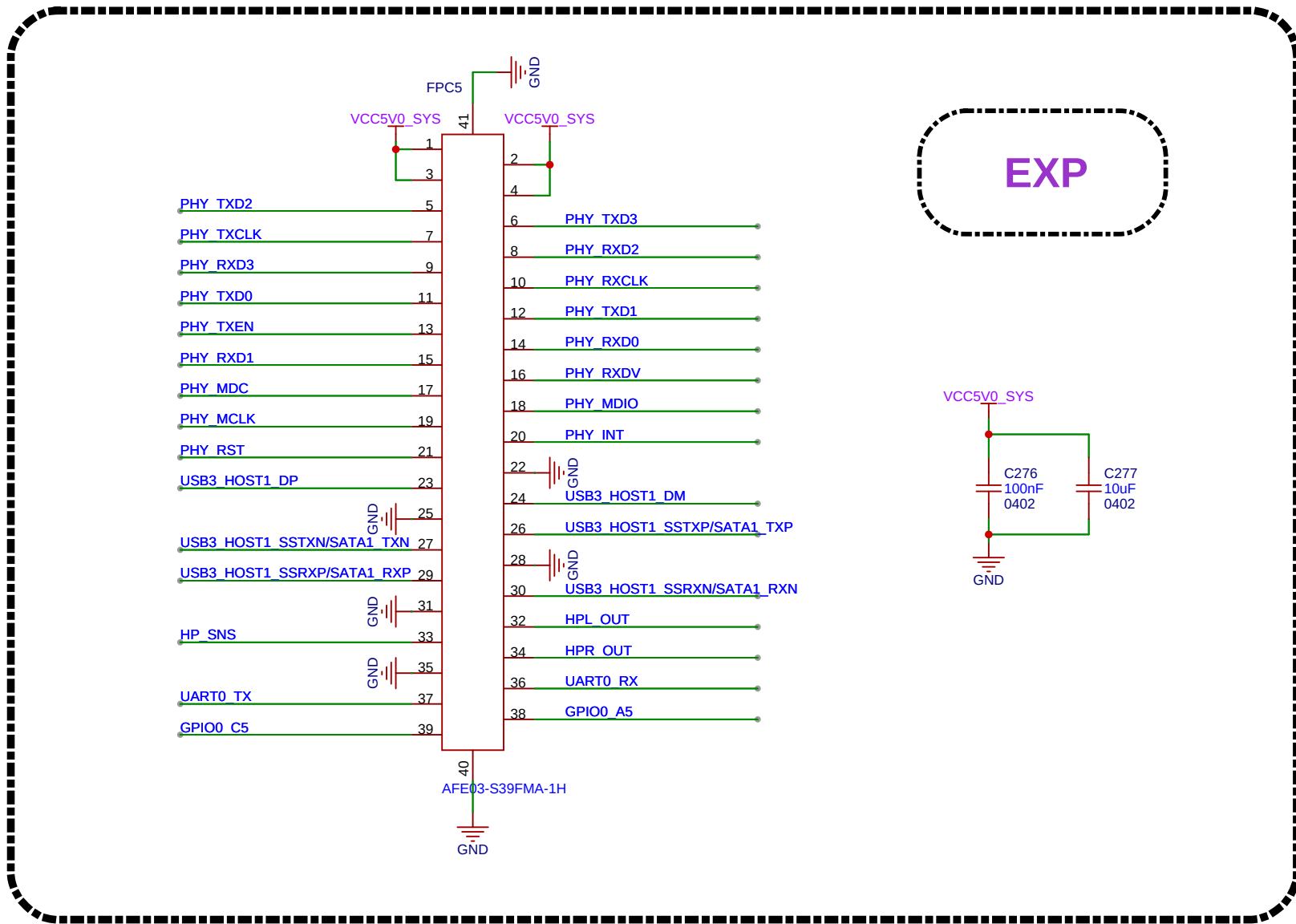
AUDIO



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